Tesla's homopolar dynamo

The disk EMF from the Lorentz force, Tesla's two-disk machine, and a cross-excited dual-motor machine with resonance, an output tank and rim bearings — CLAUDE.md §3f, M9n

kronspice simulation report · 2026-11-07 · GA: 20 generations of 30, seed 1

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Summary

Tesla's U.S. Patent 406,968 (1889) puts two conducting disks on one shaft in fields of opposite polarity, joins their rims with a flexible metal belt and takes the current from the two shafts, so no brush bears on a fast rim. This example derives the disk EMF from the Lorentz force by grade selection, builds the machine from kronspice's N machine element, compares Tesla's two-disk arrangement with a single disk and a rim brush, and evolves the two-disk machine for efficiency. A second part (the dual motor, written up in REPORT.md) removes the magnets: two disks on their own motors excite each other through each other's field coils, and the work studies resonance, an RLC output tank with a transformer and rectifier, and ball bearings as rim collectors.

- **Lorentz EMF = formula.** The EMF per disk from q(v×B) is 2.8149 V, equal to ½Bω($r_o^2 - r_i^2$) (tested to 10^{-12}).
- **Two disks beat one disk with a rim brush.** Into the same 10 mΩ load the two-disk machine gives 3044.6 W at efficiency 0.9801; the single disk with a rim brush gives 711.7 W at 0.9477. Ledger residual 0 in both.
- **Evolved machine.** r_o 0.361 m, 6.8 mm thick, 1.22 T, 1 Ω load: 24.93 V per disk, 2485 W into the load from 2486 W at the shaft, efficiency 0.9998.
- **Stacking lowers the self-excitation threshold.** Stacked 8 cm apart the dual motor self-excites at 794 rad/s (rim 119 m/s), below the 157 m/s copper hoop limit; side by side it needs 1206 rad/s (181 m/s).
- **The machine resonates without a capacitor** (shaft inertia as the second store): f_{em} = 0.587 Hz, Q = 41. At the 7.8 Hz Schumann frequency the response is forced, not resonant: ±346 A on 6.5 kA.
- **Output tank.** One R_L || C || L_p tank between the coils, with a 1 : 12.5 transformer, gives 5.34 kW to the AC load at h = 1 (scaled to 10 kA); rectified, the bank holds 449 V and feeds 5.1 kW (scaled to 10 kA).
- **Rim ball bearings** work only with superconducting field coils: a PI-regulated design holds 19.81 A at 1.60 T, 102 W to the load and 29 W in the bearings from 131.5 W at the drives. Every watt is paid by the motors; the ledgers close.

Background

A disk spinning at angular speed ω in an axial field B carries its charges at v = ω×r. The force q(v×B), the grade-1 part of <F·$Q^\dagger$> (*em.LorentzGA*; Joot, Theorem 3.16), pushes them radially, so the disk is a source of EMF between axle and rim. *em.MotionalEMF* integrates v×B along a radius and must equal

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EMF = ½ B ω (r_o² − r_i²)
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This is standard physics (CLAUDE.md §3f, the electric pair of the Lorentz force). The mechanical torque needed to keep the disk turning equals the electrical power drawn, so the energy ledger shows inflow from the shaft and outflow to the load and losses.

The model

Each disk is one N machine element with two windings, the disk's radial path and the field winding (driven at 1 A, the field strength in the coupling), and one speed-matrix entry K = ½·B·($r_o^2 - r_i^2$). The element gives the speed voltage ω·K·i_f in the disk and the torque K·i_{disk}·i_f on the shaft (Kron 1951, Epilogue eq. 9). Each shaft is a mechanical node driven at ω, so the ledger books the shaft power as inflow and shows it leaving as heat in the load, the disks, the belt and the contacts.

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Reference design	Value
Copper disks	15 cm, 5 mm thick, 1 cm axle
Field, speed	0.8 T, 3000 r/min
Load	10 mΩ
Contacts and belt (declared, illustrative, not measured)	50 μΩ shaft, 500 μΩ rim brush, 100 μΩ belt

Results: Tesla's two-disk machine

	Two disks, belt (Tesla)	Single disk, rim brush
EMF per disk, Lorentz force	2.8149 V	2.8149 V
$\frac{1}{2}B\omega(r_o^2 - r_i^2)$	2.8149 V	2.8149 V
Disk spreading resistance	1.45 μΩ	1.45 μΩ
Load	5.5178 V, 3044.6 W	2.6678 V, 711.7 W
Shaft (ledger inflow)	3106.4 W, torques 4.94 + 4.94 N·m	750.9 W, torque 2.39 N·m
Efficiency	0.9801	0.9477
Ledger residual	0	0

- The two disks' EMFs add (open circuit: 2 × EMF at the terminals, to 10^{-6}).
- The belt costs far less than a brush on a rim moving at 47 m/s, so the two-disk machine puts more than four times the power into the same load.
- **Evolved** (genes: outer radius, thickness, field up to 1.5 T, load) for at least 2 kW with the least loss, 20 generations of 30, seed 1 (562 evaluations): r_o 0.361 m, 6.8 mm thick, 1.22 T, 1 Ω load, 24.93 V per disk, 2485 W into the load from 2486 W at the shaft, efficiency 0.9998.

Results: the cross-excited dual motor

The same copper disks without magnets: each disk on its own motor, its current running through the field coil of the other unit (Rikitake's cross-excitation, as in *earth_machine*), optionally with a feedback coil carrying its own current. The coupling M (ω-EMF per ampere) is the coil's flux through the disk annulus over 2π (*em.MutualCircles* averaged over the turns), L is Nagaoka's, R the copper of a 4-turn coil in a 6 cm × 2 cm window. Liquid-metal shaft contacts (5 μΩ), the motor rotors and the coupling of two coils on one former (k = 0.95) are declared.

	Side by side	Stacked on one axis, 8 cm apart
Coupling M, own former	2.196×10^{-7} V·s/A	2.196×10^{-7} V·s/A
Coil's stray coupling to its own disk	— (a feedback coil buys it with copper)	0.519 M, free
Coil–coil mutual	—	3.77 μH (k = 0.39)
Self-excitation threshold	1206 rad/s, rim 181 m/s (K = 0); 1180 rad/s with K = 1	794 rad/s, rim 119 m/s
Against the copper hoop limit (157 m/s, 220 MPa declared)	above	below

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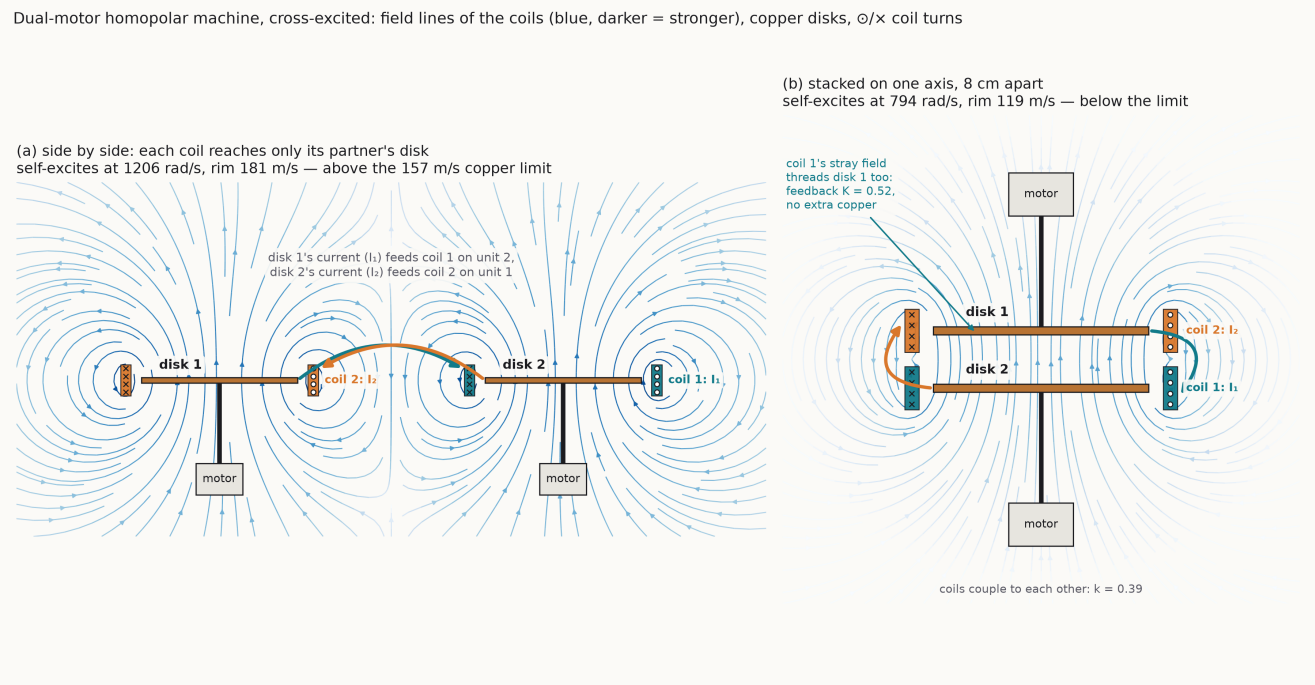


Figure 1. The two dual-motor layouts with the coils' field lines (blue, darker = stronger), copper disks and coil turns. (a) Side by side: each coil reaches only its partner's disk; threshold 1206 rad/s, rim 181 m/s, above the copper limit. (b) Stacked 8 cm apart: coil 1's stray field also threads disk 1 (built-in feedback K = 0.52) and the coils couple with k = 0.39; threshold 794 rad/s, rim 119 m/s.

- A feedback coil side by side adds coupling and its own resistance in proportion, so the threshold hardly moves (K = −0.5 raises it to 3566 rad/s).
- With both shafts held at one speed the field grows at exactly $(-R + (1+K)\cdot M\cdot\omega)/(I_{11} + I_{12})$ in both layouts (engine against exact, 10^{-3}).
- Driven by torque (20 N·m each), Rikitake's $\mu = R\cdot\sqrt{(J/(G\cdot M\cdot L))}$ is 9.5: the machine excites itself and settles to a steady 8 kA field; stacked it runs at 135 m/s on the rim, side by side at 206 m/s (past the limit).
- Reversals need $\mu \approx 1$: 1800 N·m and 4.7 MW at the shafts, 16 reversals in 11.5 s (mean 0.63 s) side by side, at a rim past the limit; with a feedback coil or stacked (both K > 0) the field locks to one polarity.
- Start-up overshoots the running speed, so a real machine starts on its speed drives. The ledger closes to 10^{-9}–10^{-6} of the drive work.

Resonance and the load

The stacked machine has a 50 μΩ load in each loop and bearing and air friction of 5×10^{-3} N·m·s (both declared). At 20 N·m it settles at $\omega^* = R/M_s$ = 944 rad/s (rim 142 m/s), 6.77 kA per loop, 2.29 kW into each load from 18.9 kW per motor. A homopolar loop alone (R and L) cannot resonate; the shafts' inertia is a capacitor on the mechanical channel, so the self-excited machine is an L–J oscillator. At the steady state the loop's net resistance is zero, so to first order only friction damps it:

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f_em = √(2·M_s·G/(L_s·J)) / 2π = 0.587 Hz,   Q = J·2πf_em/b = 41
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Oscillating speed by	Result
torque ±0.5 N·m (±2.5 %), swept	resonance at 0.575 Hz (softened from 0.587 by the drive), δω 50–70 rad/s against 1.6 at half the frequency; circuit = RK4 to 0.2 %, ledger 2×10^{-8}
torque ±0.5 / ±2 / ±5 N·m at f_{em}	harmonics of the loop current: 2nd 15 / 24 / 32 %, 3rd 2.6 / 4.3 / 7.5 %, 4th ≤ 2.5 %; no reversal. At ±5 N·m the speed peaks at 10 200 rpm, right at the copper limit
speed drives ±30 %, no capacitor	growth rate (−2.3 1/s at 0.9 ω*) independent of the oscillation frequency: no resonance, the loop is first order
speed drives ±30 %, capacitor across the load	slow oscillations (f_p ≤ about f0) lift the field because the speed spends part of each cycle above the AC onset; faster ones average out

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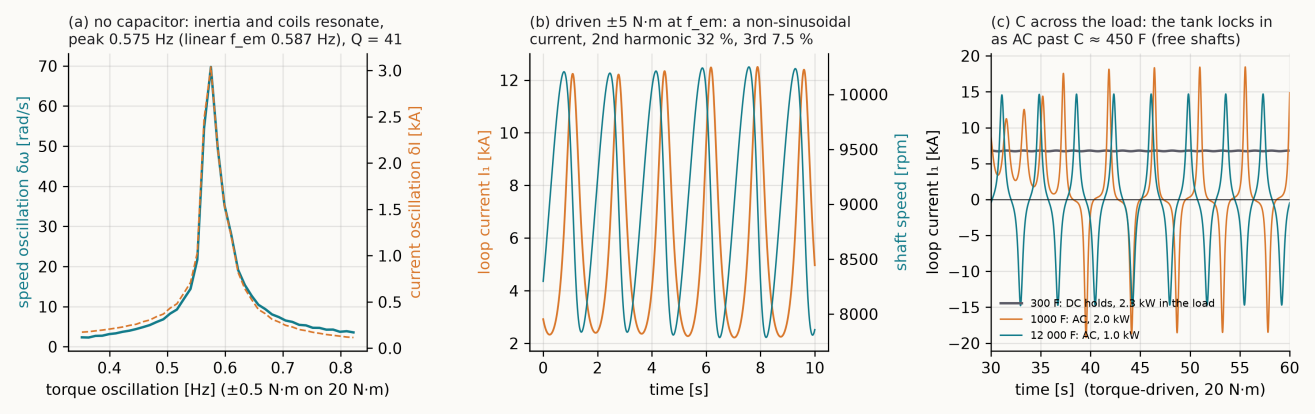


Figure 2. (a) No capacitor: speed oscillation δω and current oscillation δI against the frequency of a ±0.5 N·m torque oscillation on 20 N·m; peak 0.575 Hz (linear f_{em} 0.587 Hz), Q = 41. (b) Driven ±5 N·m at f_{em}: loop current and shaft speed against time; the current is non-sinusoidal (2nd harmonic 32 %, 3rd 7.5 %). (c) Capacitor across the load, torque-driven at 20 N·m: 300 F stays DC (2.3 kW in the load), 1000 F and 12 000 F lock in as AC (2.0 kW and 1.0 kW).

With the shafts held at speed the tank locks in once $C > L/R_L^2$ (5401 F); with free, torque-driven shafts the inertia joins the tank and the DC state gives way at C ≈ 454 F (bisection, from 1 % off the DC state).

C across each load	tank f0	Q	outcome	loop current	load
30 F	7.9 Hz	0.07	DC	6.77 kA	2.29 kW
300 F	2.5 Hz	0.24	DC	6.77 kA	2.29 kW
1000 F	1.37 Hz	0.43	AC	±7.6 kA at 0.22 Hz, 3rd harmonic 66 %	2.03 kW
3000 F	0.79 Hz	0.75	AC	±8.0 kA at 0.23 Hz, 3rd 59 %	1.83 kW
12 000 F	0.40 Hz	1.49	AC	±8.5 kA at 0.27 Hz, 3rd 37 %	1.00 kW
24 000 F	0.28 Hz	2.11	AC	±7.3 kA at 0.14 Hz, 3rd 78 %	0.67 kW

- Locked in, the machine is an AC generator below both the tank and f_{em}; it gives the load less, because much of the AC circulates between the coils and the capacitor. Circuit = RK4 to 0.6 %, ledger 10^{-8}.
- A faster speed swing does not let the capacitor shrink: $Q = R_L \cdot \sqrt{(C/L)}$ falls as $\sqrt{C}$. With the rim-safe swing h = 0.16 only 3000 F grows (+0.17 1/s at f0/4).
- Ceilings for this copper machine: tank f0 ≤ 0.99 Hz with the speed held (Q ≥ 1), ≤ 3.4 Hz with free shafts (Q ≥ 0.29); with no capacitor, f_{em} is 0.59 / 1.9 / 5.9 Hz at 20 / 200 / 2000 N·m.
- **Shafts oscillated at 7.8 Hz** (first Schumann frequency): ±101 rad/s about ω* takes ±274 N·m per shaft and gives a forced ±346 A at 7.8 Hz on the 6.5 kA DC, exactly $M_s \cdot I \cdot \delta\omega/(2\pi f \cdot L_s)$; harmonics 1.4 % and 0.5 %; the load gets only 3 W at 7.8 Hz. Resonance at 7.8 Hz would need a 31 F tank (Q = 0.07) or 3.5 kN·m, 3.3 MW per shaft.
- As a Schumann source the coils are a 282 A·m² dipole at 7.8 Hz, above the natural Schumann field (about 1 pT) only within 300 m, radiating 6×10^{-22} W: a local 7.8 Hz magnet, not a Schumann source.

rpm = 1768·(1 ± h·sin 2π·7.8t)

The speed drives impose the profile. For a self-excited loop at a held speed

```
I(t) ~ exp((M_s·ω0 − R)/L_s · t) · exp(−β·cos 2πft),   β = M_s·ω0·h / (L_s·2πf)    (~ : proportional to)
```

so the mean speed decides growth and the sine only ripples the field, with harmonics $2 \cdot I_n(\beta)/I_0(\beta)$ (modified Bessel functions).

- **Air coils:** the machine self-excites at 9014 rpm, so at 1768 rpm the field decays at −18.7 1/s for h = 0, 0.5 and 1 alike.
- **Iron magnetic circuit** (gain k = 5.10, declared, chosen so 1768 rpm is the onset): L_s = 69 µH, mean field 1.6 T at 10.5 kA; ripple 0.0093, 0.047 and 0.093 of the DC at h = 0.1, 0.5 and 1; RK4 and circuit equal the exact value to 3×10^{-4}, ledger 3×10^{-9}.
- **Antiphase** shafts: the field decays slightly (−0.04 1/s at h = 1). A 6 F or 24 F capacitor (Q = 0.015–0.03) changes nothing.

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- Drives: ±50 N·m at h = 0.1, ±503 N·m at h = 1; up to 186 kW is swapped with the rotor every cycle, so the drives must be four-quadrant (regenerative).

h	DC	AC at 7.8 Hz	at 15.6 Hz	load DC	load AC	AC share
0.1	10 kA	93 A peak (66 A rms)	0.2 A	5.0 kW	0.2 W	0.004 %
0.5	10 kA	466 A (330 A rms)	5.4 A	5.0 kW	5.4 W	0.11 %
1	10 kA	932 A (659 A rms)	22 A	5.0 kW	22 W	0.43 %

One RLC tank between the two machines' coils, transformer and rectifier

One current runs shaft 1 → disk 1 → R_{int} → coil 1 → R_L || C || L_p → coil 2 → R_{int} → disk 2 → shaft 2 → common. Shaft 2 counter-rotates and coil 2 is wound to aid coil 1; co-rotating, the EMFs cancel and the current decays as a plain R–L loop (test: −1.0 1/s, matching the formula to 5 %). This is a series self-excited dynamo; the two-loop Rikitake mode, reversals included, is gone. The iron gain is 4.45, set so the DC onset is 1768 rpm. At 7.8 Hz each machine is a source with R_s = 0.275 mΩ behind X_s = 2.945 mΩ (L_s = 60 µH), and the output is conjugate-matched to it with tank Q = 3.

Element	Per machine	The one tank between the coils
R_L	63.7 mΩ	127 mΩ
C	8.79 F	4.40 F
L_p (primary)	216.5 µH	433 µH
Transformer	1 : 12.5, k = 0.98	1 : 12.5, k = 0.98
AC load R_{ac}	10 Ω	20 Ω

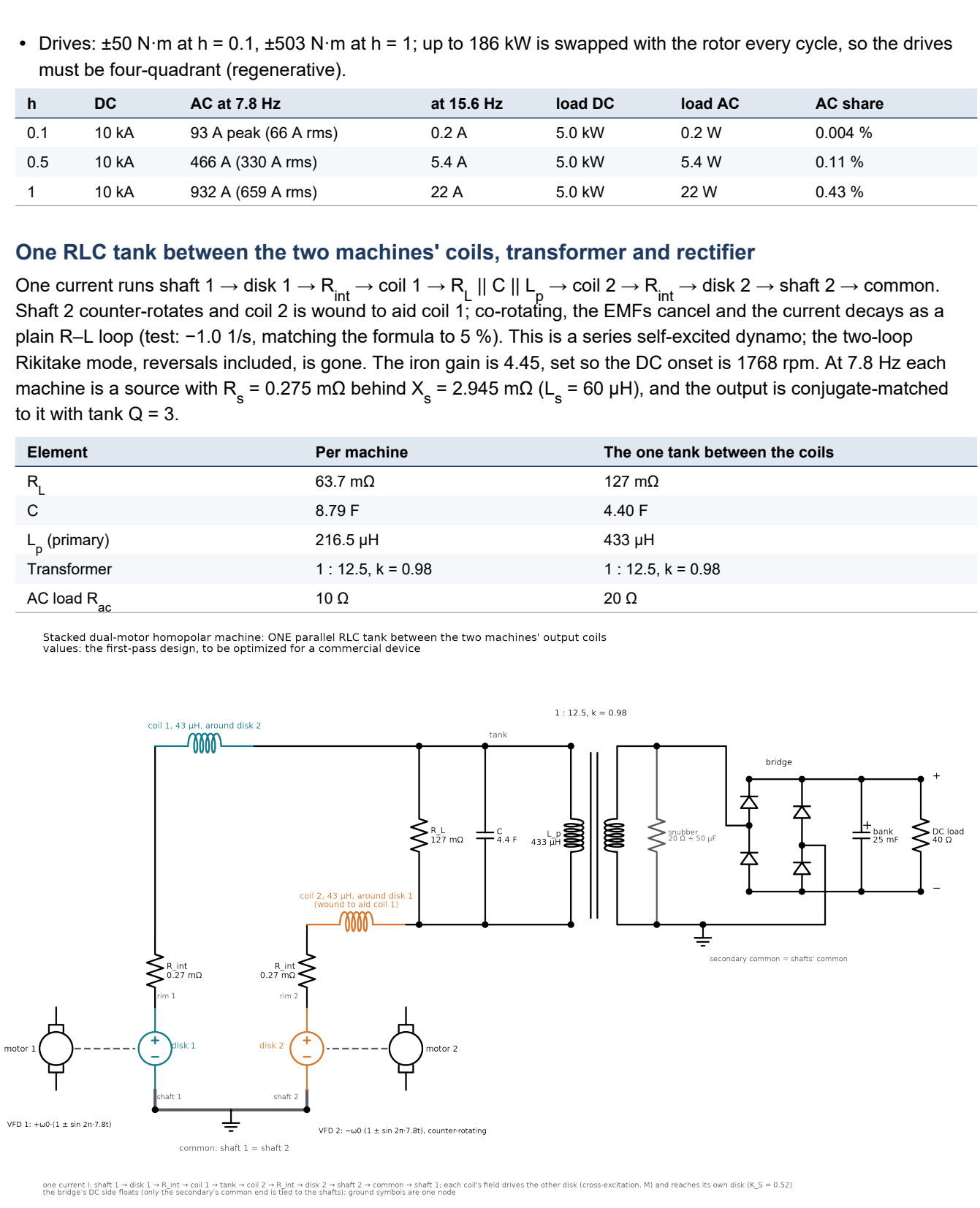


Figure 3. Schematic of the stacked dual-motor machine with one parallel RLC tank between the two machines' output coils: motors and speed drives (counter-rotating), disks, R_{int} 0.27 mΩ, coils 43 µH each, the tank R_L 127 mΩ || C 4.4 F || L_p 433 µH, a 1 : 12.5 transformer (k = 0.98), snubber, bridge, 25 mF bank and 40 Ω DC load. The secondary's common end is tied to the shafts' common. Values are a first-pass design.

Parametric pumping: with the matched output the 7.8 Hz speed sine pumps the field, so it holds below the DC onset. Scaled to a 10 kA loop current:

h	steady mean rpm	loop AC	primary AC	R_L	secondary	R_{ac}
0.1	1758	1043 A	295 A	150 W	76 V / 3.8 A	144 W

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h	steady mean rpm	loop AC	primary AC	R_L	secondary	R_{ac}
0.5	1592	4306 A	1218 A	2.55 kW	316 V / 15.8 A	2.45 kW
1.0	1337	6359 A	1799 A	5.56 kW	470 V / 23.5 A	5.34 kW

The circulating current rises with the tank Q, but the power that crosses the transformer does not:

Tank Q	L_p	C	primary AC	secondary	AC power out
1	1299 µH	3.75 F	659 A	23.0 A	10.8 kW
3	433 µH	4.40 F	1799 A	23.1 A	10.9 kW
10	130 µH	6.64 F	5833 A	22.8 A	10.6 kW
30	43.3 µH	13.1 F	15 176 A	19.8 A	8.0 kW

- The circulating part is the primary's magnetizing current and stays in the tank; only the in-phase part reaches the secondary, at most $EMF^2/(8 \cdot R_s)$ per machine.
- **Rectifier into a bank** (Q = 3 design, one bridge, 25 mF bank pre-charged to 400 V, 40 Ω DC load, x = 0.756, h = 1): at 8.8 kA loop current the bank holds 396 V with 33 V p-p ripple at 15.6 Hz and the DC load takes 3.9 kW; scaled to 10 kA, 449 V and 5.1 kW. Ledger closes to 4×10^{-8}.
- Circuit = RK4 (21 666 A against 21 669 A), ledger 10^{-6} of the drive work; the primary carries the loop's 10 kA DC (21.7 kJ at 433 µH), so its core needs an air gap or a cancelled DC flux.

Ball bearings at the rims and an adjustable feedback

Each disk's rim current goes through a ball-bearing bank (P element): measured contact drop 0.73 V per bank (Safdarzadeh et al., *Lubricants* 12, 148, 2024); damage limit 0.3 A/mm² of Hertzian contact area scaled to a 300 mm rim (a 7.5× extrapolation, labelled); Huber κψ never measured, so swept. The declared bank (four 300 mm races at 5 kN) carries 24.8 A at the damage limit. Because the bearing drop is a fixed voltage, at fixed feedback the loop has one unstable equilibrium current $I^* = 2 \cdot V_b/(2 \cdot M_s \cdot \omega - R_{loop})$.

- **Copper machine:** no equilibrium at 1.0 or 1.5 × onset, I^* = 29.5 kA at 2.0 and 4.5 kA at 2.5. At 10 kA the bearings would carry 404× their damage limit (15 kW of heat). 1.6 T costs 37 kW per copper coil against 82 W generated per disk at the bearing limit.
- **Bearing-matched design:** superconducting field coils of 2438 turns (loop inductance 45 H), set point 19.8 A, load 0.261 Ω; each disk makes 3.32 V, each bank drops 0.73 V. At fixed feedback 5 % off the equilibrium the current runs away to 17.19 A or 22.53 A after 600 s (RK4: 17.14 A and 22.48 A).
- **PI regulator** on the adjustable feedback K_v: from a 7.4 A start pulse it reaches 95 % in 283 s and holds 19.81 A at $K_v = 0$, 1.60 T; load 102 W, bearings 29 W, drives 131.5 W. Books: drive 393 kJ = load 300 kJ + bearings 86 kJ + stored 7.5 kJ (residual 2×10^{-8}).
- **Huber torque:** in EM mode the drives pay the same 0.355 N·m per shaft at κψ = 0, 10^{-4} and 10^{-3} (at 10^{-2} the current collapses); thermal mode returns at most η = 10 % of the bearing heat (≤ 0.008 N·m).

Ledger and validation

- Tests: Lorentz EMF = formula to 10^{-12}; ledger shaft power = Σ torque·ω to 10^{-9}; torque per disk = K·I with $I = 2 \cdot EMF/R_{total}$; ledger residual below 10^{-12}.
- Dual motor: threshold growth = exact in both layouts; circuit = RK4 with feedback coils and stacked; the one tank equals the per-machine tanks (10^{-9}); $I_2 = -I_1$ at every step.
- Every watt at the outputs comes from the two motors and their speed drives; the neutrino and other unmodeled channels are zero. No step adds energy: not the stacking, the resonance, the tank Q nor the transformer.

Limits of the model

- Contact, belt and brush resistances, shaft contacts, rotor inertia, windage, hoop strength, iron gain, transformer, snubber, diode, bank and load values are declared, illustrative inputs, not measured. The component values are a first pass, to be optimized later for a commercial device.

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- The rim-bearing damage limit is a Hertz-scaled extrapolation; the cryocooler power of the superconducting coils is outside the model; the iron is linear (no saturation to stop a runaway).
- The bridge is nonlinear, so only the engine models it; its check is the ledger.

Reproducing the results

```
go run ./examples/homopolar [-gens 20] [-pop 30] [-p 2000]
go run ./examples/homopolar -dual      # all dual-motor parts, A–I
go run ./examples/homopolar -rim       # rim bearings only
go test ./examples/homopolar
python examples/homopolar/viz/make_report.py examples/homopolar/homopolar_report.pdf
```

Sources: N. Tesla, U.S. Patent 406,968 (1889); P. Joot, *Geometric Algebra for Electrical Engineers*, Theorem 3.16; G. Kron, *Equivalent Circuits of Electric Machinery* (1951), Epilogue eq. 9; doc/M9n.md. The full dual-motor write-up is REPORT.md in this folder.

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